

Alberto Maria Mongardini

Curriculum Vitae

Current Position

09/2025 – **PostDoctoral Researcher**, *Department of Applied Mathematics and Computer Science, Cybersecurity Engineering*, Technical University of Denmark (DTU), Lyngby, Denmark.
Current

Previous Work Experience

06/2024 – **PostDoctoral Researcher**, *Department of Computer Science*, Sapienza University of Rome, Rome, Italy.
08/2025
04/2023 – **Short-Term Scholar**, *George Mason University*, Department of Computer Science and Department of Cyber Security Engineering, Fairfax, USA.
10/2023
Host: Prof. Giuseppe Ateniese.

Education

2020 – 2024 **PhD in Computer Science**, *Sapienza University of Rome*, Italy.
Advisor: Prof. Alessandro Mei, **Cycle:** XXXVI, **Grade:** Summa cum Laude
PhD Schools:

- *International Winter School of Northern Lights Deep Learning Conference 2023*, Tromsø, Norway, 2023
- *International Summer School on Deep Learning*, Gran Canaria, Spain, 2021
- *CS Doctoral Workshop*, Bertinoro, 2021
- *Challenges in Building Billion User Cloud Applications*, Rocca Sinibalda, 2019

2019 – 2020 **Master degree in Computer Science**, *Sapienza University of Rome*, Italy, **Advisors:** Alessandro Mei.
Grade: Summa cum Laude

2019 **Challenges in Building Billion User Cloud Applications**, Rocca Sinibalda, Italy.

2015 – 2018 **Bachelor degree in Computer Science**, *Sapienza University of Rome*, Italy, **Advisor:** Prof. Alessandro Mei, **Grade:** Summa cum Laude.

2010–2014 **Classical education degree**, *Liceo classico Lucrezio Caro*, **Grade:** 100/100.

Grants

2023 **Travel grant**, *43th IEEE International Conference on Distributed Computing Systems (ICDCS 2023)* Travel Grant, IEEE Computer Society's Technical Community on Distributed Processing, ≈ \$1'000.

- 2023 **Avvio Alla Ricerca Tipo 2**, *Beyond Traditional Stealth Addresses: Empowering Bitcoin Transactions with Enhanced Expressiveness and Privacy*, Sapienza University of Rome.
Role: **Principal Investigator** $\approx$ 2'000€
- 2023 **Grant for young researchers**, *Study and Detection of NFT Wash Trading on the Ethereum Blockchain*, Bando mobilità internazionale PHD 2022, Sapienza University of Rome.
Role: **Principal Investigator** $\approx$ 3'600€
- 2022 **Avvio Alla Ricerca Tipo 1**, *NFTs Washtrading Detection within the Ethereum ecosystem*, Sapienza Università di Roma.
Role: **Principal Investigator** $\approx$ 1'000€
- 2021 **Progetti di Ricerca Grandi**, *Prebunking: predicting and mitigating coordinated inauthentic behaviors in social media*, Sapienza Università di Roma.
Role: **Component** $\approx$ 40'000€

Scholarships and Awards

- 2026 **Seal Of Excellence (Score: 95.8/100)**, *European Commission*, Marie Curie Postdoctoral Fellowships 2025.
- 2025 **Postdoctoral Research Fellowship**, *Technical University of Denmark*, Copenhagen, Denmark, Winner of a one-year and four months research fellowship at the Department of Cybersecurity of the Technical University of Denmark..
- 2025 **Research Collaborator**, *Sapienza University of Rome*, Rome, Italy.
Awarded an independent research contract to conduct research on market manipulations involving meme coins at the Department of Computer Science.
Code: ICE 04/2025
- 2024 **Postdoctoral Research Fellowship**, *Sapienza University of Rome*, Rome, Italy, Winner of a one-year research fellowship at the Department of Computer Science of Sapienza University of Rome.
Code: AR-B 05/2024.
- 2020 **Research Collaborator**, *Sapienza University of Rome*, Rome, Italy.
Awarded an independent research contract for the ARCHEM project: Development of neural networks for the recognition of museum artworks, at the Department of Computer Science.
Code: ICE 14/2021
- 2020 **Doctoral scholarship**, *Winner of a three-year doctoral scholarship at the Department of Computer Science of Sapienza University of Rome, cycle XXXVI*, Dipartimento di Informatica, Sapienza Università di Roma.

Academic services

- 2026 **T-IFS**, *Reviewer for the IEEE Transactions on Information Forensics and Security*.
- 2025 **T-IFS**, *Reviewer for the IEEE Transactions on Information Forensics and Security*.
- 2025 **ICWSM 2025**, *Reviewer at 19TH International AAAI Conference on Web and Social Media*.
- 2025 **ICWS 2025**, *Reviewer at IEEE International Conference on Web Services*.

- 2024 **ICWSM 2024**, *Reviewer at 18TH International AAAI Conference on Web and Social Media*.
- 2023 **The Web Conference 2023**, *Reviewer at the Web Conference 2023*.

Talks and Seminars

- 2025 **Conference Speaker**, TORONTO, CANADA, Speaker of the paper: "TGDataset: Collecting and Exploring the Largest Telegram Channels Dataset" for KDD '25.
- 2025 **Seminar**, ROME, ITALY, Cybersecurity Master's Program, Sapienza University of Rome, Speaker of a seminar for students in the Master's Degree in Cybersecurity program, "Cryptocurrency Market Manipulations: Uncovering the Blockchain Wild West".
- 2025 **Guest Lecturer**, ROME, ITALY, Distributed Systems Master's Program, Sapienza University of Rome, Delivered lectures on wash trading activities on Ethereum within the course held by Prof. Alessandro Mei.
- 2024 **Guest Lecturer**, ROME, ITALY, Distributed Systems Master's Program, Sapienza University of Rome, Delivered lectures on detection of scams and frauds in the Telegram ecosystem within the course held by Prof. Alessandro Mei.
- 2023 **Conference Speaker**, HONG KONG, CHINA, Speaker of the paper: "A Game of NFTs: Characterizing NFT Wash Trading in the Ethereum Blockchain" for ICDCS '23.

Student Supervision

- Ongoing **M.Sc. Student**, *Technical University of Denmark (DTU)*, Student: Emil Nymark Trangeled, Topic: Honeywords generators with LLMs.
Cosupervised with Prof. Emmanouil Vasilomanolakis
- Ongoing **M.Sc. Student**, *Sapienza University of Rome*, Student: Alessandro Angeleri, Topic: Detection of disposable accounts in market manipulations.
Cosupervised with Prof. Massimo La Morgia
- Ongoing **Erasmus+ Student**, *Technical University of Denmark (DTU)*, Student: Stelios Malamas, Topic: Development of HoneyPersonas.
Cosupervised with Prof. Emmanouil Vasilomanolakis

Conference Participation

- 2025 **KDD '25**, *Toronto, Canada*.
- 2024 **Devcon 2024**, *Bangkok, Thailand*.
- 2024 **CyberTech Europe**, *Rome, Italy*.
- 2023 **ICDCS '23**, *Hong Kong, China*.
- 2022 **International Conference on emerging Networking EXperiments and Technologies (CoNEXT)**, *Rome, Italy*.

International Collaborations

Active research collaborations with leading international research groups and scientists:

- **Prof. Sushil Jajodia** (George Mason University, USA) – *Fellow IEEE, Fellow ACM*. Collaboration on cyber deception and IP theft defense mechanisms.
- **Prof. Giuseppe Ateniese** (George Mason University, USA) – *Fellow IEEE*. Collaboration on Bitcoin accountability and privacy-preserving protocols.
- **Prof. Jie Wu** (Temple University, USA) – *Fellow IEEE, Fellow AAAS*. Collaboration on social network security and fake engagement detection on Telegram.
- **Prof. Emmanouil Vasilomanolakis** (Technical University of Denmark - DTU). Collaboration on honeypots, AI-driven deception, and threat intelligence.

Research Software and Open Science

- Artifact **Code & Dataset: Meme Coin Market Manipulations.**
- **Zenodo:** zenodo.org/records/17830944
 - **Description:** Comprehensive replication package for the cross-chain analysis of 34,988 meme coins (Ethereum, BSC, Solana, Base). Includes the full longitudinal dataset, Dune Analytics SQL queries, and Python detection logic for identifying artificial growth strategies (Wash Trading, Liquidity Pool-Based Price Inflation - LPI) and profit extraction schemes (Rug Pulls, Pump and Dumps).
 - **Impact:** Official artifact supporting the findings presented in **USENIX Security 2026** ("A Midsummer Meme's Dream").
- Dataset **TGDataset: Code & Data for Telegram Threat Research.**
- **GitHub:** github.com/SystemsLab-Sapienza/TGDataset
 - **Zenodo:** [10.5281/zenodo.7640712](https://zenodo.org/records/10.5281/zenodo.7640712)
 - **Description:** The largest public dataset of Telegram channels (120k+ channels, 400M+ messages), developed to enable systematic research on deception, misinformation, and fraud. Includes preprocessing scripts and JSON-based schema documentation.
 - **Impact:** Published at **KDD 2025**. Used in peer-reviewed publications including **ICWS '23** and **TWEB** (for fake channel detection) and **USENIX Security 2025** (analyzing conspiracy monetization).
- Dataset **Conspiracy Dataset on Telegram.**
- **GitHub:** github.com/SystemsLab-Sapienza/conspiracy-dataset-telegram
 - **Description:** A curated list of channels involved in conspiracy narratives and monetization schemes. Contains categorization by conspiracy type and metadata (subscriber counts, message frequency).
 - **Impact:** Supports the analysis presented in **USENIX Security 2025** ("The Conspiracy Money Machine").
- Artifact **Regex for Detecting Anti-Sniping Mechanisms.**
- **Zenodo:** [10.5281/zenodo.7604918](https://zenodo.org/records/10.5281/zenodo.7604918)
 - **Description:** Curated collection of regular expressions designed to detect smart contract features that prevent sniper bot attacks (cooldowns, transfer limits, blacklists).
 - **Impact:** Employed in studies investigating DeFi market manipulation (**CAWW '23** and **ACM TOIT**).

Technological Transfer

2019 – 2021 **Technology Transfer Specialist**, *SystemsLab*, Department of Computer Science, Sapienza University of Rome.

Contributed to industrial innovation and development projects focused on applying AI and AR to real-world scenarios.

- **ARCHEM (Augmented Reality for Cultural Heritage in Etruria Meridionale):**

- *Context:* Competitive project funded by Regione Lazio (POR FESR).
- *Contribution:* Designed and deployed **Computer Vision models** (based on YOLO and ResNet architectures) for the real-time recognition of museum artworks on mobile devices. Optimized Neural Networks for on-device inference using PyTorch and TorchScript.

- **Airmapp (Airport Indoor Navigation):**

- *Context:* Industrial collaboration for the development of an AR-based indoor navigation system.
- *Contribution:* Engineered the scalable **Cloud Backend** infrastructure (Google App Engine, Firebase) to support high-concurrency user requests and manage geospatial data for indoor positioning.

Research Media Coverage

- 2026 **"Why so many meme coins fail almost immediately?"**, *Fast Company*, Leading US business & innovation magazine.
- 2024 **"De Rafapal a La Quinta Columna: así se financian los canales de conspiraciones en Telegram?"**, *El Pais*, National daily.
- 2024 **"Telegram: The Modern Hades of the Digital Underworld"**, *Security Distillery*, Blog.

Publications

- USENIX 2026 **USENIX Security Symposium**, *A Midsummer Meme's Dream: Investigating Market Manipulations in the Meme Coin Ecosystem*, **Alberto Maria Mongardini**, and Alessandro Mei.
- ACNS 2026 **International Conference on Applied Cryptography and Network Security**, *Stealth and Beyond: Attribute-Driven Accountability in Bitcoin Transactions*, **Alberto Maria Mongardini**, Daniele Friolo, Giuseppe Ateniese.
- KDD 2025 **ACM SIGKDD Conference on Knowledge Discovery and Data Mining**, *TG-Dataset: Collecting and Exploring the Largest Telegram Channels Dataset*, Massimo La Morgia, Alessandro Mei, **Alberto Maria Mongardini**.
- USENIX 2025 **USENIX Security Symposium**, *The Conspiracy Money Machine: Uncovering Telegram's Conspiracy Channels and their Profit Model*, Vincenzo Imperati, Massimo La Morgia, Alessandro Mei, **Alberto Maria Mongardini**, and Francesco Sassi.
- TOIT **ACM Transactions on Internet Technology**, *The Blockchain Warfare: Investigating the Ecosystem of Sniper Bots on Ethereum and BNB Smart Chain*, Federico Cernerà, Massimo La Morgia, Alessandro Mei, **Alberto Maria Mongardini**, and Francesco Sassi.
- T-IFS **IEEE Transactions on Information Forensics and Security**, *DARD: Deceptive Approaches for Robust Defense against IP Theft*, **Alberto Maria Mongardini**, Massimo La Morgia, Sushil Jajodia, Luigi Vincenzo Mancini, and Alessandro Mei.

- IEEE **IEEE Access**, *Hard to See, Harder to Block: Exploring the Challenges in Email*
ACCESS *Tracking Pixels Detection*, Federico Cernera, Vincenzo Imperati, Massimo La Morgia,
Alessandro Mei, **Alberto Maria Mongardini**, Eugenio Nerio Nemmi, Luciano
Ortenzi, and Francesco Sassi.
- TWEB **ACM Transactions on the Web**, *Pretending to be a VIP! Characterization and*
Detection of Fake and Clone Channels on Telegram, Massimo La Morgia, Alessandro
Mei, **Alberto Maria Mongardini**, and Jie Wu.
- ICWS 2023 **IEEE International Conference on Web Services - IEEE ICWS '23**, *It's a*
Trap! Detection and Analysis of Fake and Clone Channels on Telegram, Massimo
La Morgia, Alessandro Mei, **Alberto Maria Mongardini**, and Jie Wu.
- ICDCS 2023 **IEEE 43th International Conference on Distributed Computing Systems -**
ICDCS '23, *A Game of NFTs: Characterizing NFT Wash Trading in the Ethereum*
Blockchain, Massimo La Morgia, Alessandro Mei, **Alberto Maria Mongardini**, and
Eugenio Nemmi.
- CAAW 2023 **Companion Proceedings of the Web Conference - CAAW '23**, *Ready, Aim,*
Snipe! Analysis of Sniper Bots and their Impact on the DeFi Ecosystem, Federico
Cernera, Massimo La Morgia, Alessandro Mei, **Alberto Maria Mongardini**, and
Francesco Sassi.